pfi

Python for Fixed Income

COURSE 5 · ADVANCED MACHINE LEARNING

Case Study: Bagging and the Spread Cross-Section

WEEK 1 · DAY 5

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Today: a forest for a number

Morning: two hands-on sections

5.1 Regression trees and forests on the benchmark bonds. 5.2 The threshold rule, moved onto the training rows.

Afternoon: a case study

OLS, a depth-4 tree, and a 300-tree forest on the same bonds, the same folds, the same metric.

Does an average of trees describe the spread cross-section better than Course 4's straight line?

The answer depends on which columns the models get, and on how the bonds are split.

How a regression forest works

The problem	A bond's spread is a number, so a leaf must answer with a number
One tree	Ask the question that leaves the smallest squared errors; a leaf gives its bonds' mean log spread
The forest	Bootstrap sample of bonds per tree, some columns per split, deep trees, answers averaged
You control	n_estimators, max_features, and min_samples_leaf
What comes out	A fitted log spread per bond, turned back into basis points to read

Regression trees and forests

A leaf gives the mean log spread of its training bonds. The depth-4 tree asks score <= 9.5 first; its 16 leaves give 16 fitted spreads to 615 bonds. The first bond's leaf holds 56 of them.

0.8816

oob_score_ of a regressor: R-squared on log_oas

41.0 bp

MAE out of bag

16.2 bp

MAE on the bonds each tree was fitted on

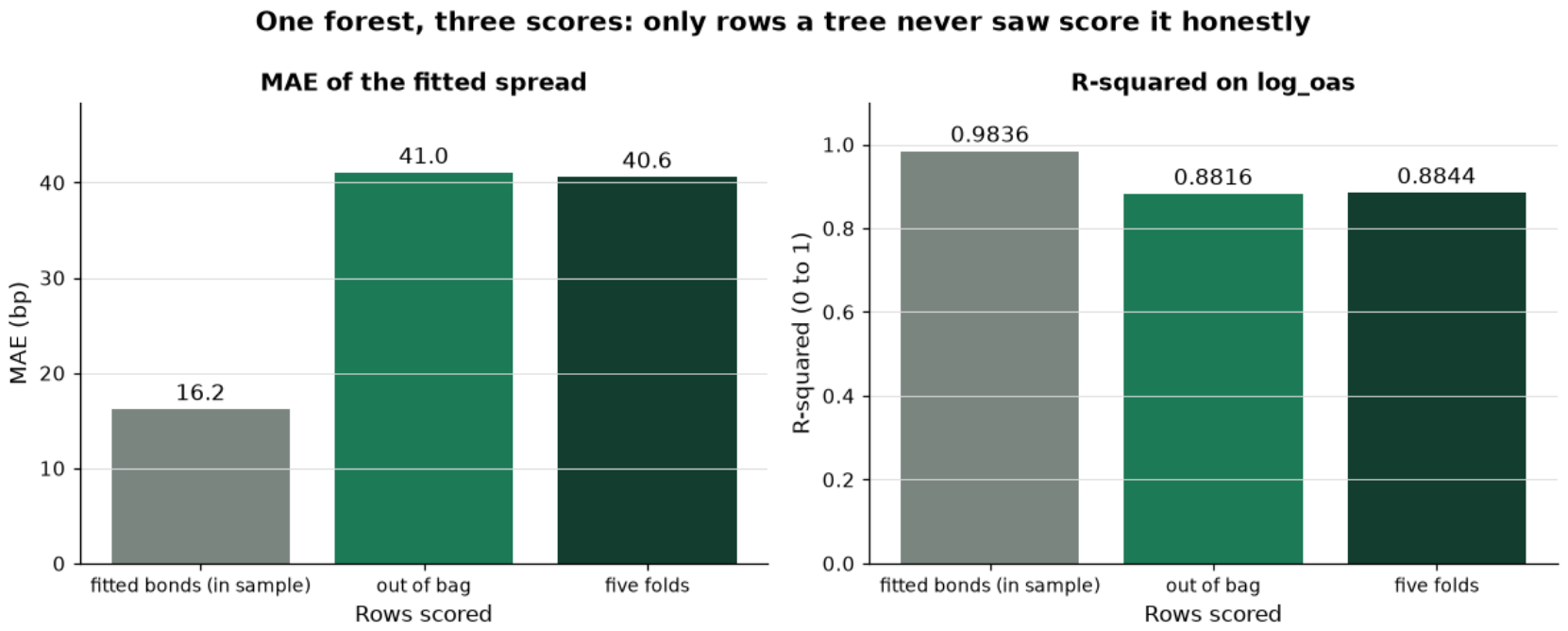
40.6 bp

MAE from five folds

```
RandomForestRegressor(n_estimators=300, oob_score=True, random_state=SEED, n_jobs=1)
```

Day 2: oob_score_ of a classifier is accuracy. Same name, different measure: check it.

One forest, three scores



In sample 16.2 bp, R-squared 0.9836. Out of bag 41.0 bp, 0.8816. Five folds 40.6 bp, 0.8844. Only rows a tree never saw score it honestly.

How out-of-fold thresholds work

The problem	A threshold must be chosen on rows the model did not fit, and not on the validation rows
Folds	Cut the training rows into five; fit a fresh copy on four, record probabilities for the fifth
One each	Every training row gets one probability from a model that never saw it
The rule	Take the highest threshold whose out-of-fold recall is at least the target
What comes out	One threshold, which the validation rows then test once

The threshold rule, on the training rows

1. Five folds

Each training row gets a probability from a model that never saw it.

2. Choose

The highest threshold with out-of-fold recall of at least 0.70.

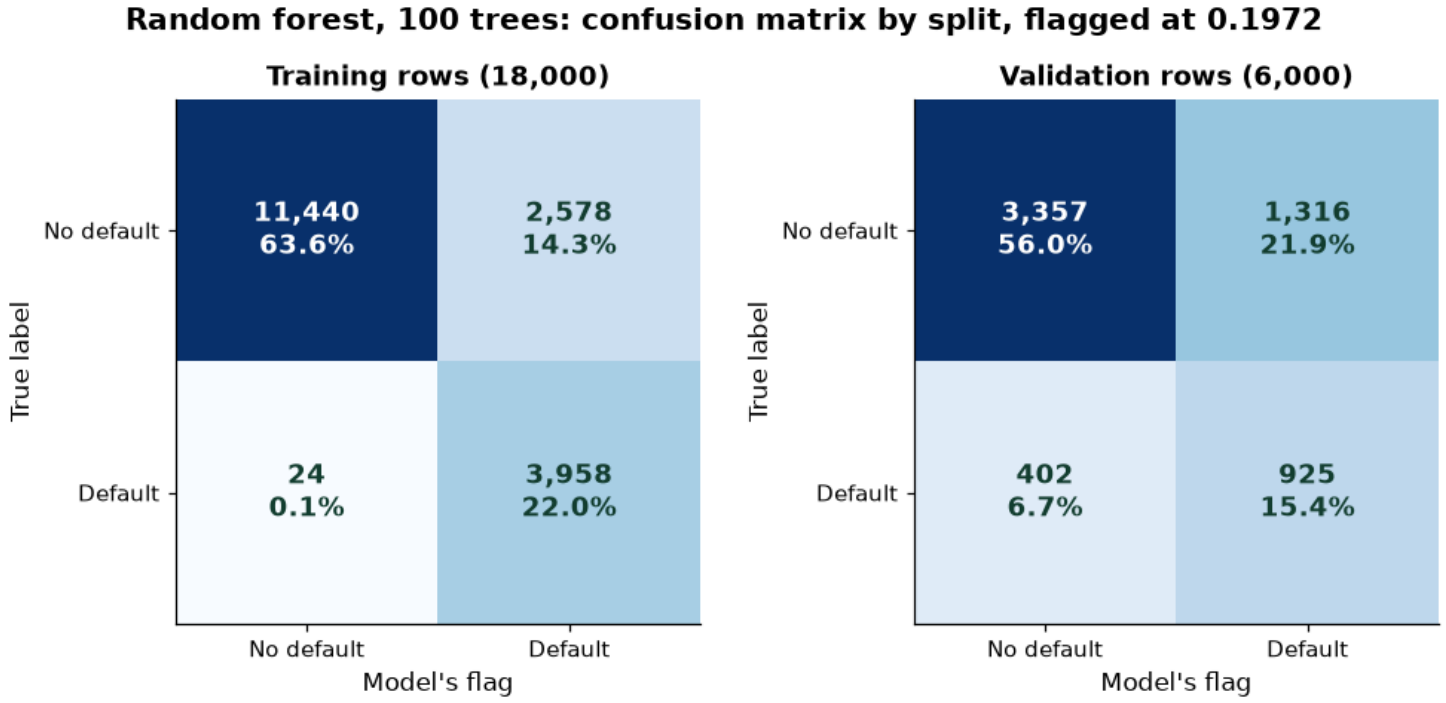
3. Report

Fit on all training rows; flag the validation rows at that threshold.

Card file	Threshold	Out-of-fold recall	Validation recall	Validation precision
Logistic regression	0.2047	0.7002	0.6956	0.3319
Forest, 100 trees	0.1972	0.7002	0.6971	0.4128

Each model gets its own threshold. Validation recall lands near 0.70, not on it: an honest number.

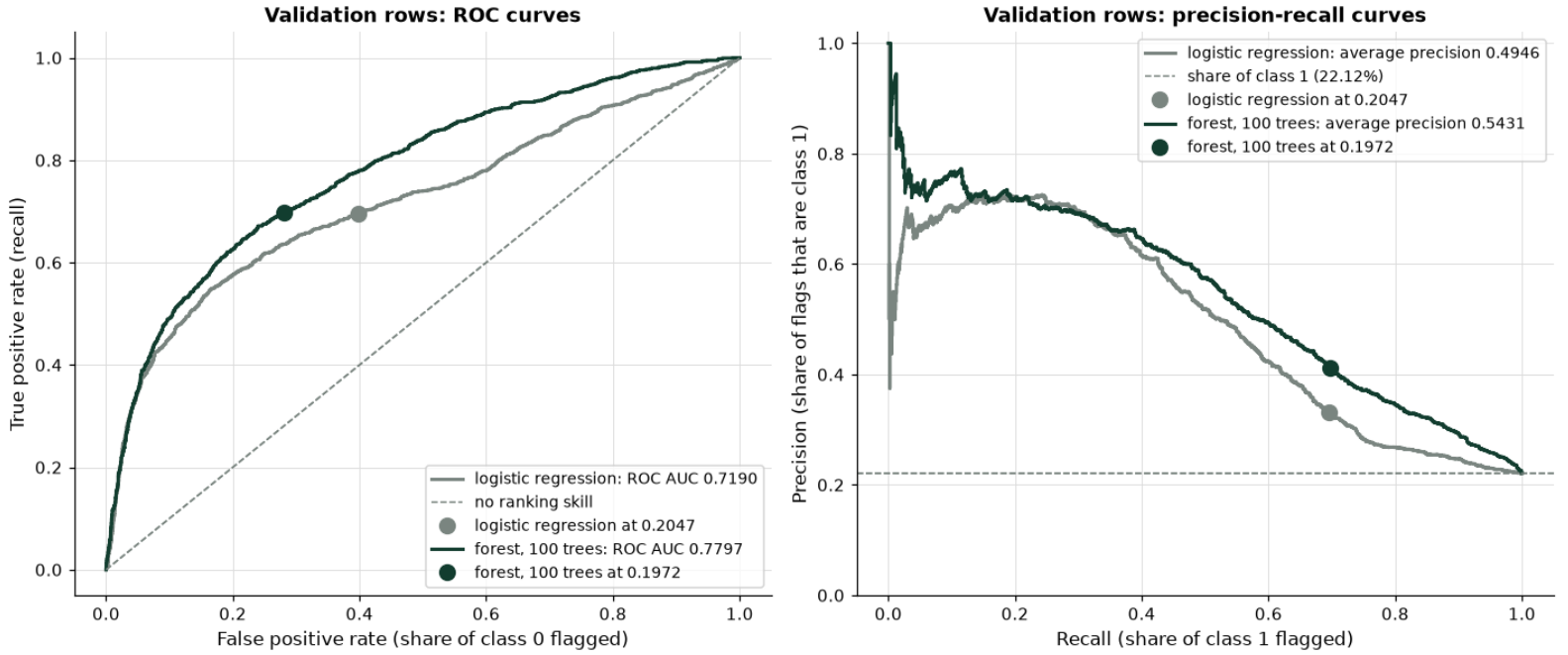
Training rows flatter the forest



Rows	Recall	Precision
Logistic, training	0.7039	0.3337
Logistic, validation	0.6956	0.3319
Forest, training	0.9940	0.6056
Forest, validation	0.6971	0.4128

Each model flagged at its own out-of-fold threshold. The forest's trees were fitted on the training rows; the threshold was not chosen on them.

Two models on one chart



Validation rows. Logistic regression: ROC AUC 0.7190, average precision 0.4946, threshold 0.2047. Forest: 0.7797 and 0.5431, threshold 0.1972. The forest's ROC curve is higher at 97 percent of false positive rates.

Case study: the spread cross-section

Three models, Course 4's 615 training bonds, the same five folds, cross-validated MAE in bp. Without and with coupon; at random and by issuer.

OLS: Course 4's pipeline

Rating and sector as indicator columns; the numbers scaled.

Trees: the rating as its number

score from 1 (AAA) to 18 (CCC), sector as indicators, the numbers as they are.

Allowed names

fitted spread

the model's spread for these characteristics

residual

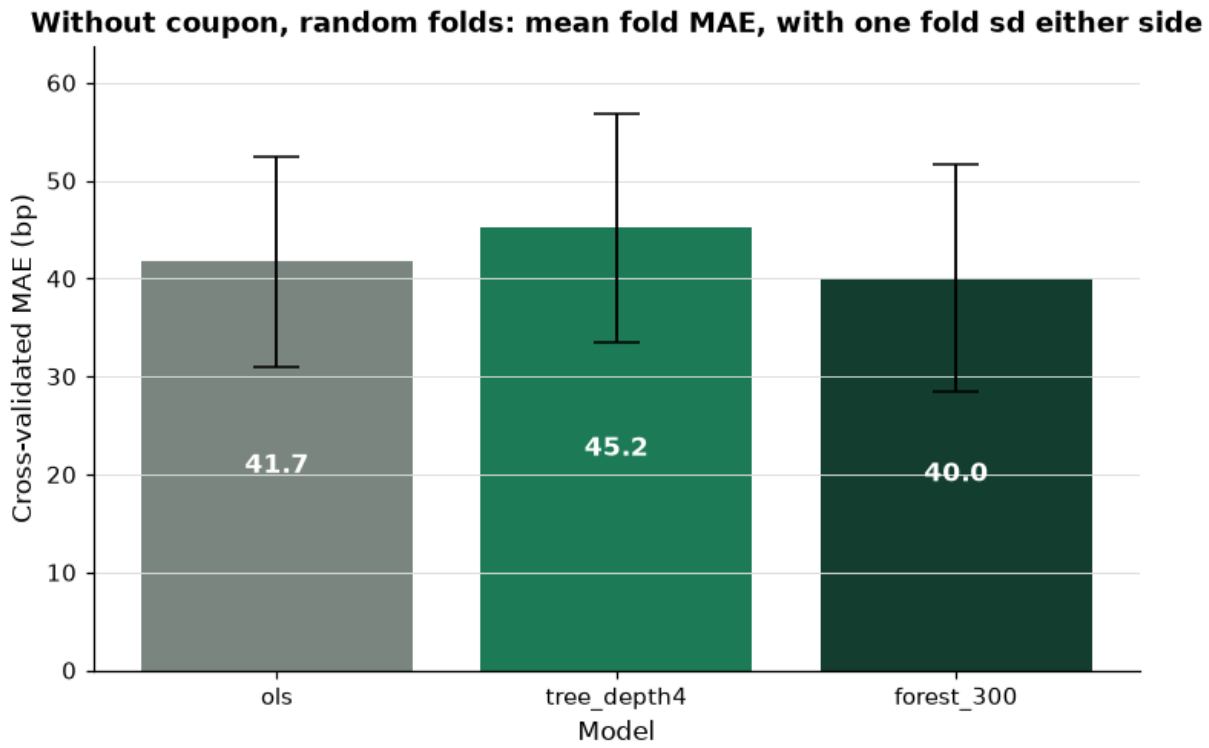
Never called

fair value or fair spread

cheap, rich, mispriced

a signal or a trade

Three models without coupon



Mean of five fold MAEs, bp

Forest 40.0, OLS 41.7, one depth-4 tree 45.2

Read the fold sd first

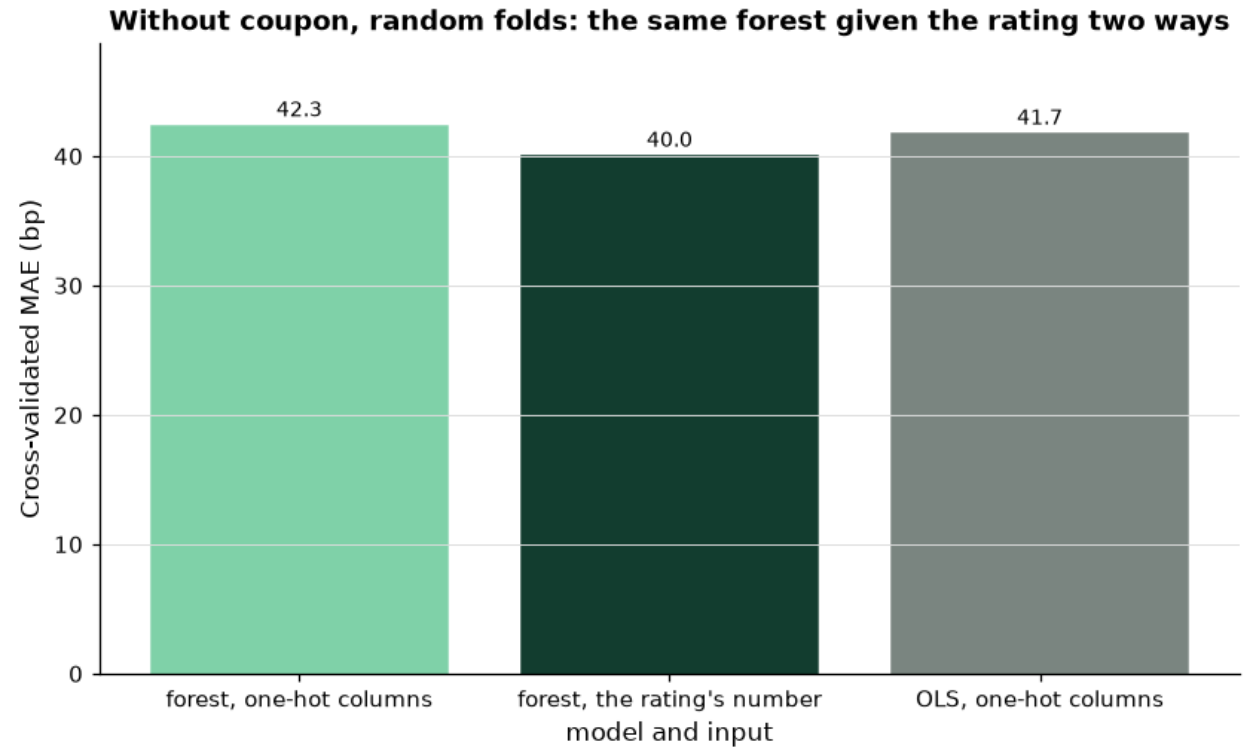
Fold sd 11.6 bp for the forest, 10.7 bp for OLS: some folds hold harder bonds.

Then compare fold by fold

On the same bonds, the forest misses by less in 3 of 5 folds.

Averaging trees beats one tree by far, and the straight line by a little.

What the forest is given



42.3 bp

forest, one-hot columns (29)

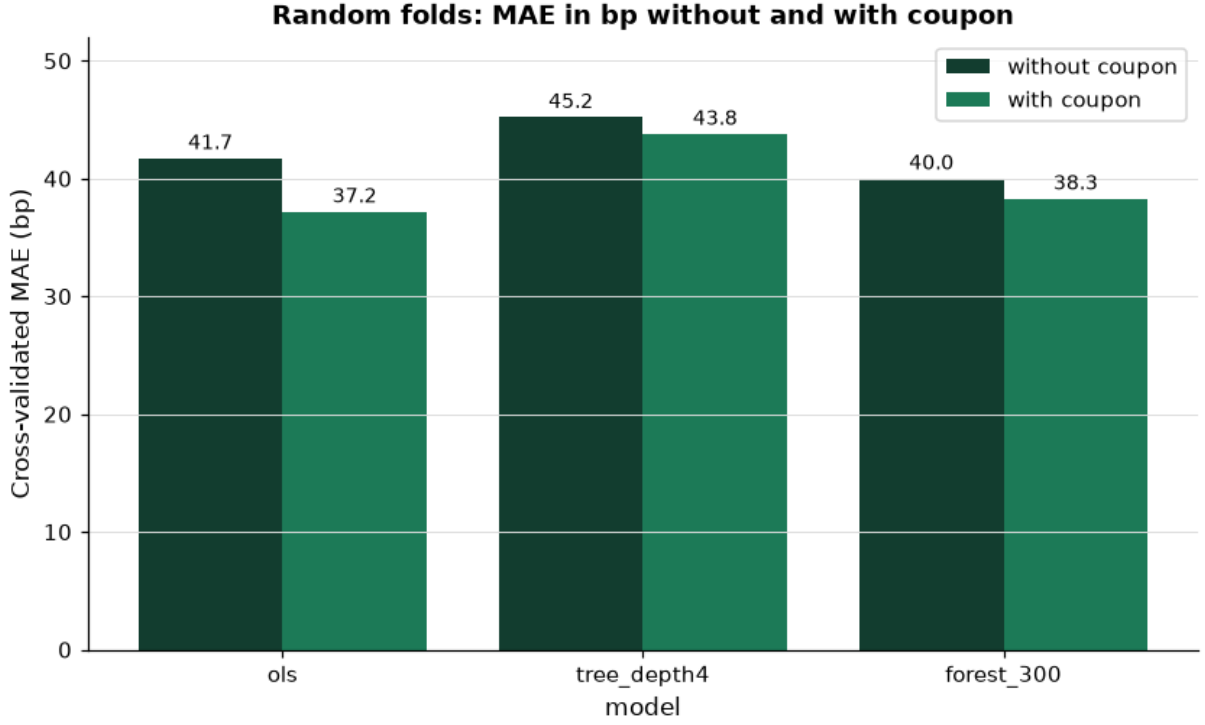
40.0 bp

forest, the rating's number (14)

A tree spends one question per rating level when the rating comes as indicators; the rating's number lets it ask about many levels at once.

What a model is given matters as much as which model it is.

Adding coupon



MAE bp	Without	With	Gain
ols	41.7	37.2	4.5
tree_depth4	45.2	43.8	1.4
forest_300	40.0	38.3	1.7

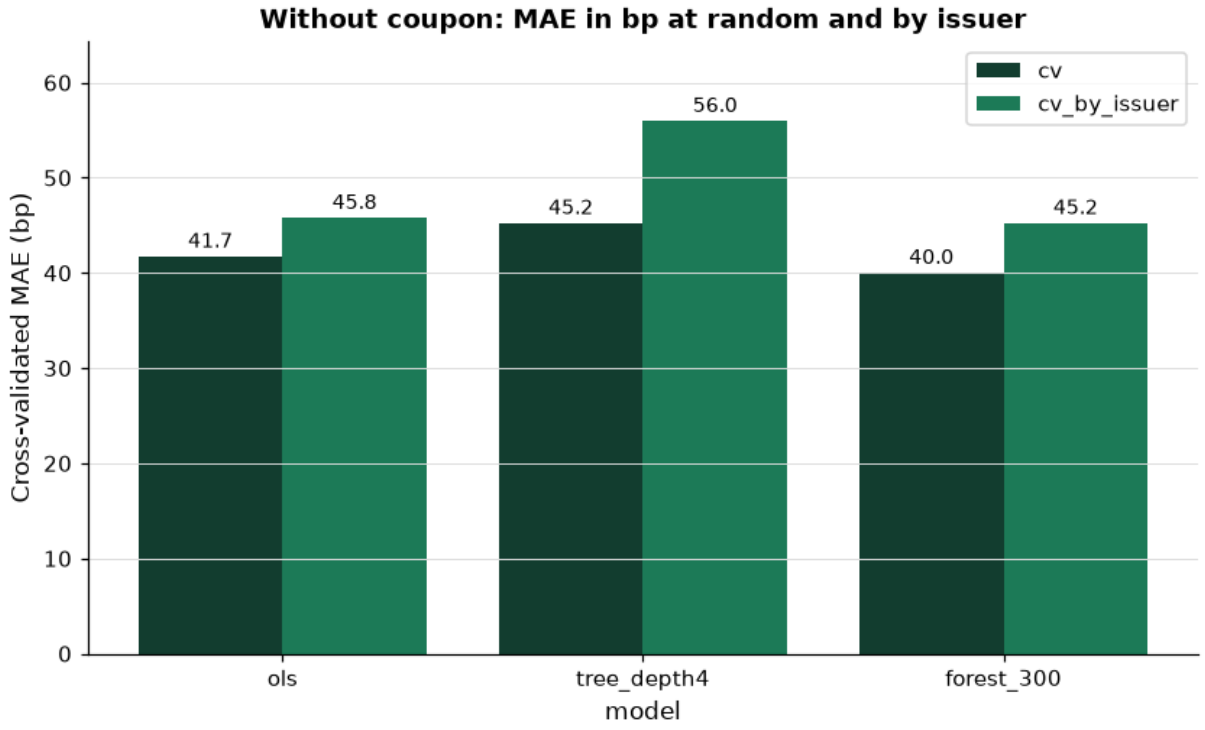
The generator sets the coupon from the yield, and the yield holds the bond's own spread: coupon partly restates the target (Course 4 day 5).

With coupon, the straight line misses by less than the forest.

How GroupKFold works

The problem	Random folds put an issuer's bonds on both sides, so a bond is scored by a model fitted on its siblings
Groups	Give every bond a group label: the issuer's ticker
Whole groups	Each issuer sits in one fold; fit on four folds' issuers, score the fifth's
You control	The number of folds and the grouping column
What comes out	Folds like KFold, scoring a model on issuers it has not seen

Splitting by issuer

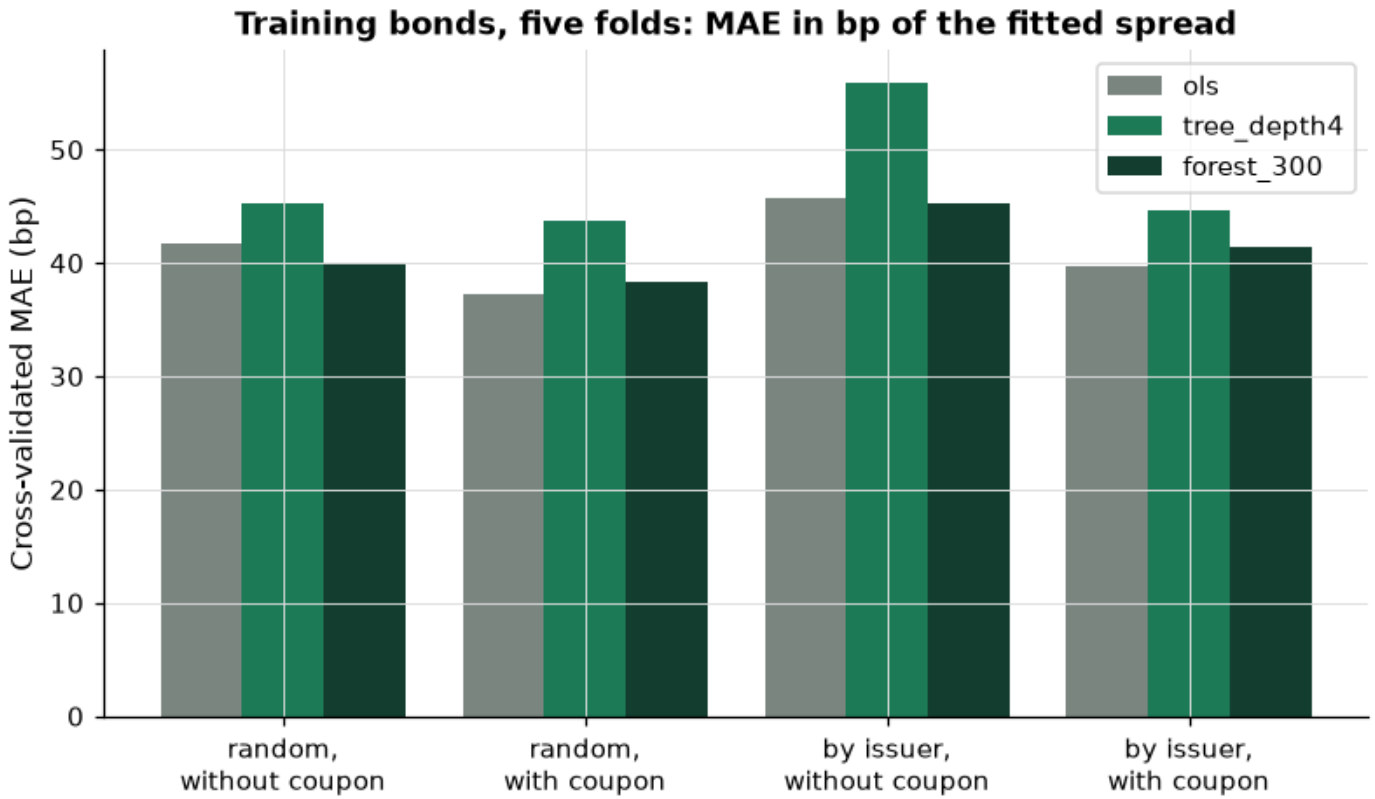


MAE bp	Random	By issuer	Cost
ols	41.7	45.8	4.0
tree_depth4	45.2	56.0	10.7
forest_300	40.0	45.2	5.2

GroupKFold(5) with groups=ticker keeps each issuer's bonds in one fold. 6 warnings: ratings held by one issuer only are unknown to OLS's encoder when that issuer is scored.

At random: a bond of an issuer the model has seen. By issuer: an issuer it has not.

The comparison



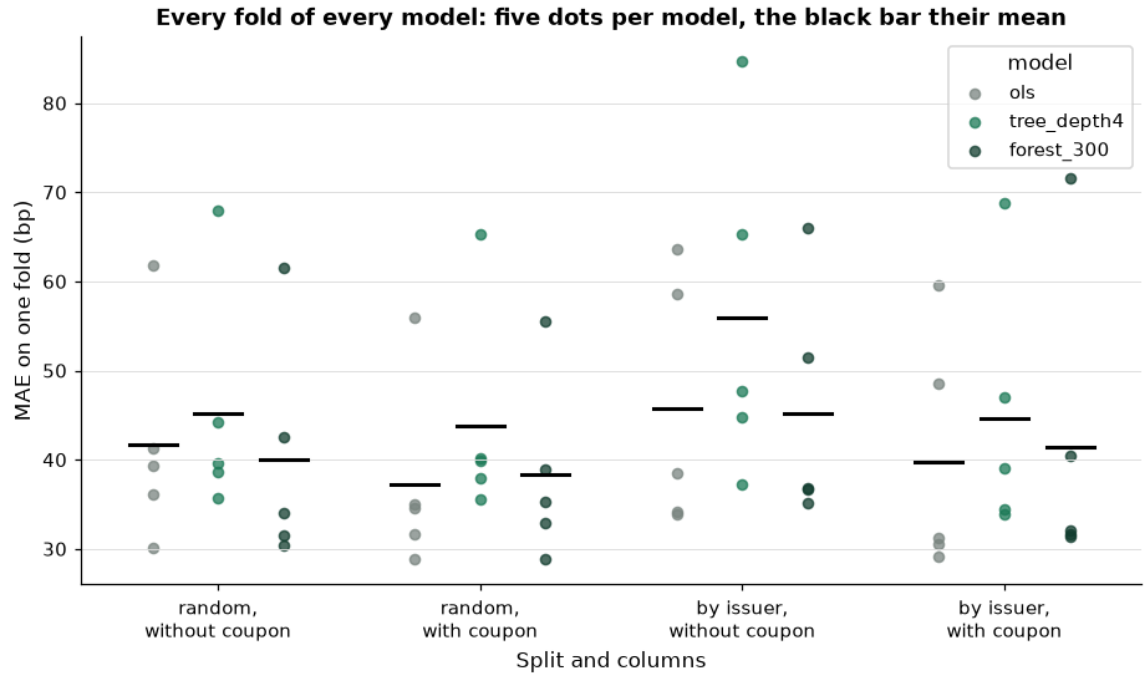
Without coupon

The forest is lowest, at random and, only just, by issuer.

With coupon

OLS is lowest under both splits. The depth-4 tree is behind everywhere.

Every fold of every model



26.7 bp

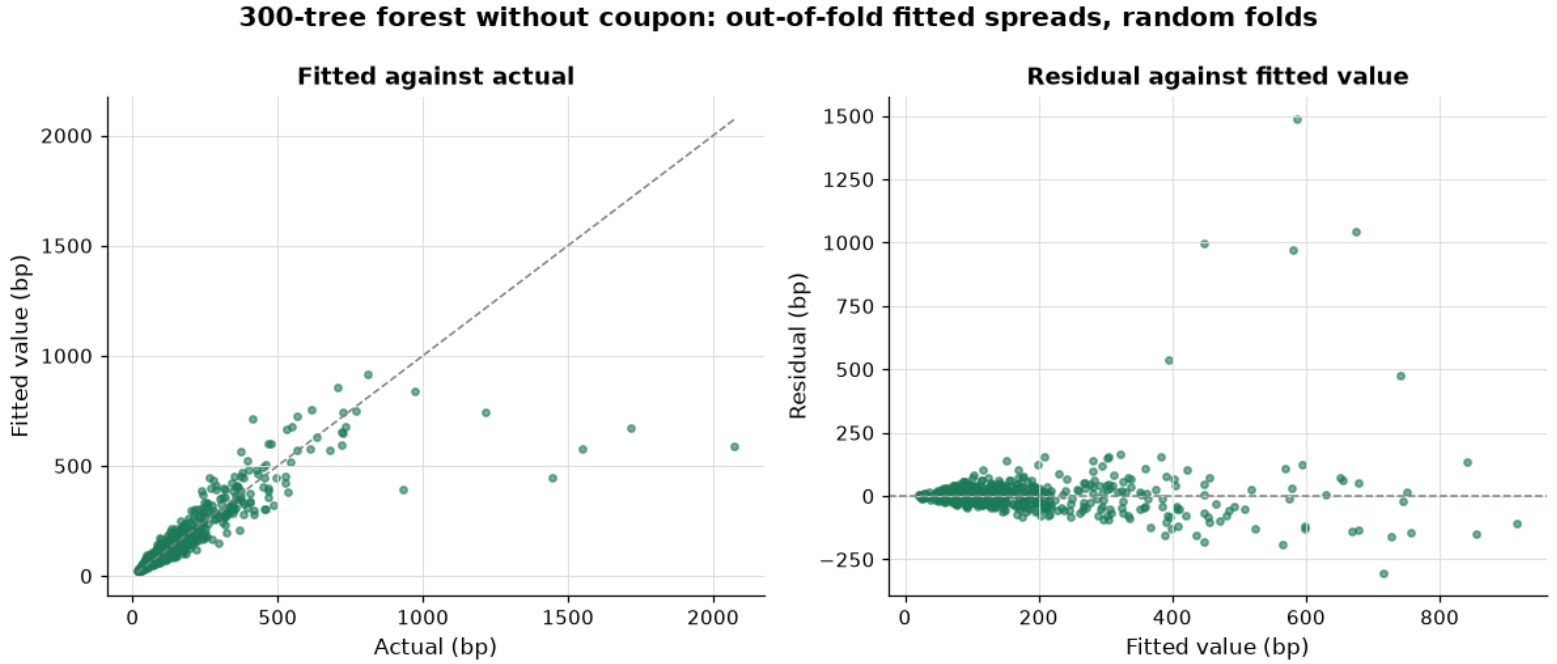
the narrowest spread of one model's five folds

1.7 bp

the widest gap between OLS and the forest

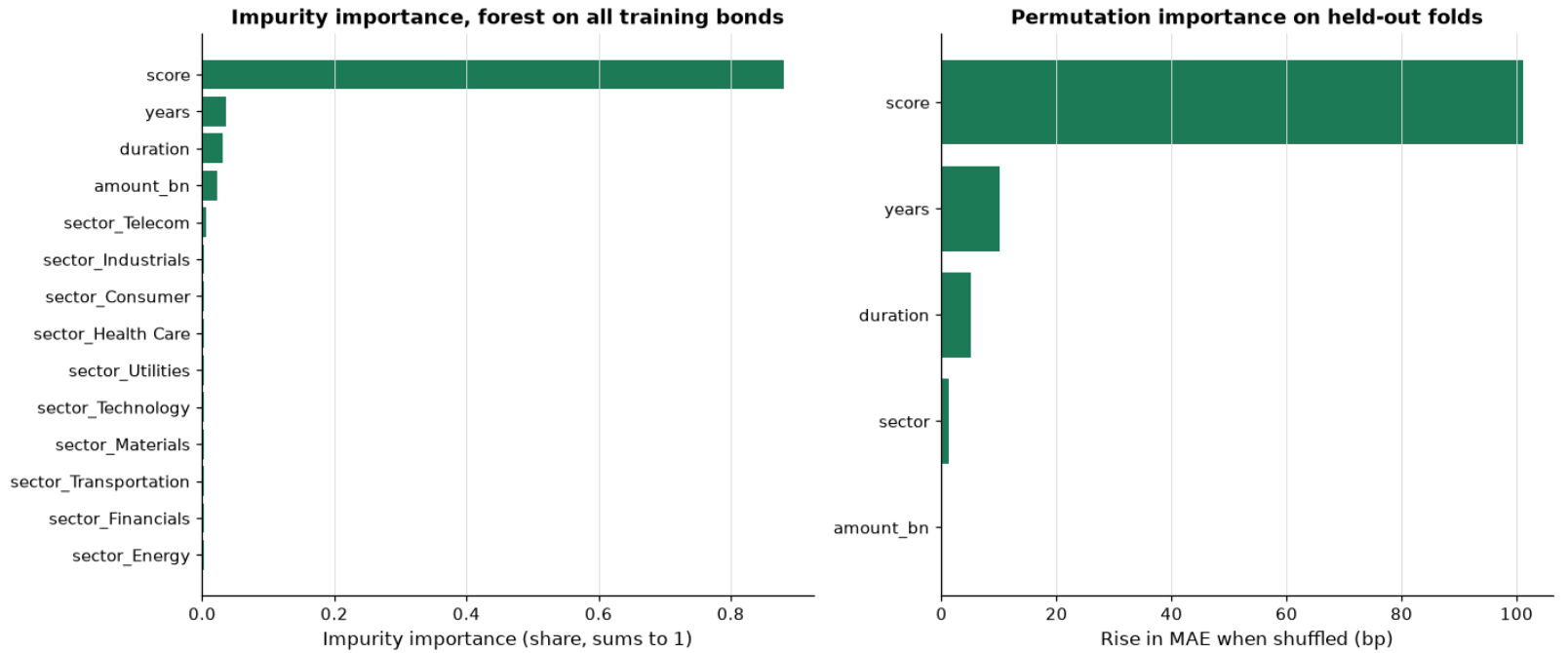
The closing opens with every result: twelve rows in one formatted table in the notebook, and this chart. A gap this size reads only in pairs.

Where the forest misses



7 bonds lie above the forest's widest fitted spread, 915.0 bp. Above the median fitted spread the forest misses by 64.4 bp on average; below it, by 15.6 bp.

What the forest leans on



The rating's number: 0.881 of the impurity, 101.2 bp when shuffled. Years 10.0 bp, duration 5.1 bp. Sector: 0.029 of the impurity, 1.2 bp. An association in this file.

What the model shows, and where it fails

What it shows

Forest 40.0 bp, OLS 41.7 bp, one tree 45.2 bp without coupon; with coupon, the line is ahead.

New issuers	Every model misses by more when the issuer was never seen.
Wide spreads	Forest MAE 79.9 bp on high yield, 19.2 bp on investment grade.
The edges	Widest OAS 2,074 bp; the forest's widest fitted spread 915.0 bp, OLS 1025.9 bp.
The data	Synthetic bonds, one date. A fitted spread is never a fair value.

Every setting of RandomForestRegressor

Setting	Default	Tune first?
n_estimators	100	
max_depth	None	
min_samples_leaf	1	tune first
max_features	1.0	tune first
bootstrap	True	
max_samples	None	tune first
criterion	'squared_error'	
oob_score	False	
ccp_alpha	0.0	

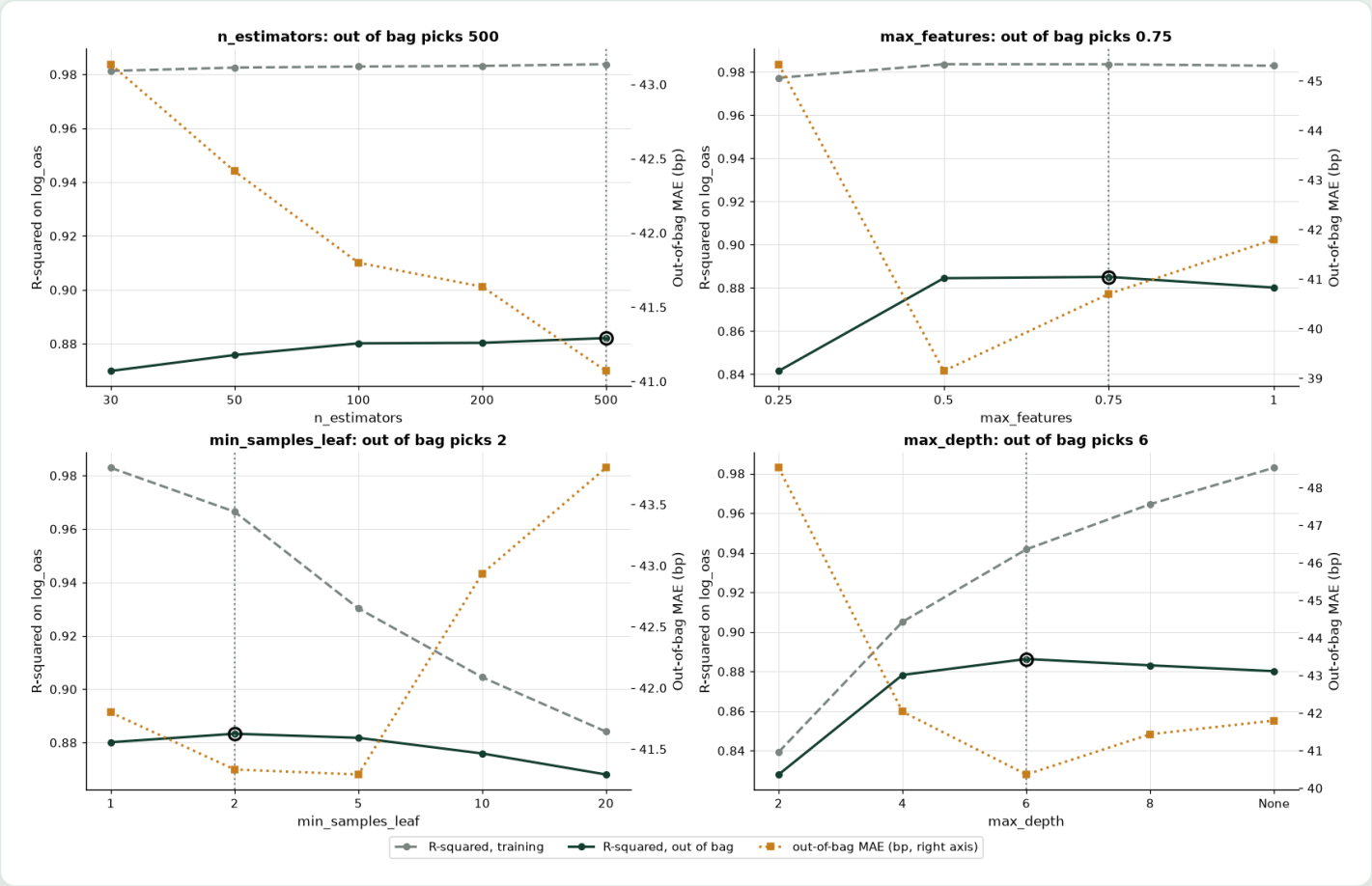
Against the classifier

No class_weight. criterion is squared_error, not gini. max_features is 1.0 (all the columns), not sqrt, so the case study's forest is bagging in day 3's sense.

The three to tune first

min_samples_leaf, max_features, and max_samples. A leaf's output is the mean log spread of its bonds, so a bigger leaf averages more bonds.

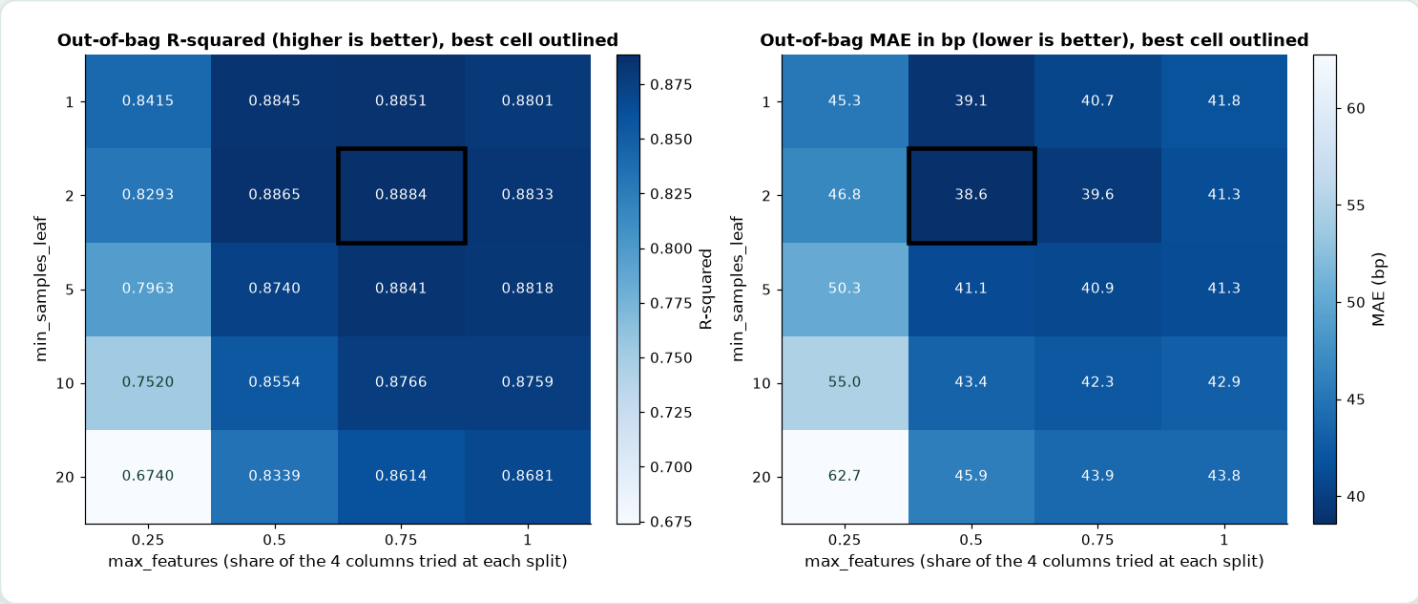
Four settings, one at a time, chosen out of bag



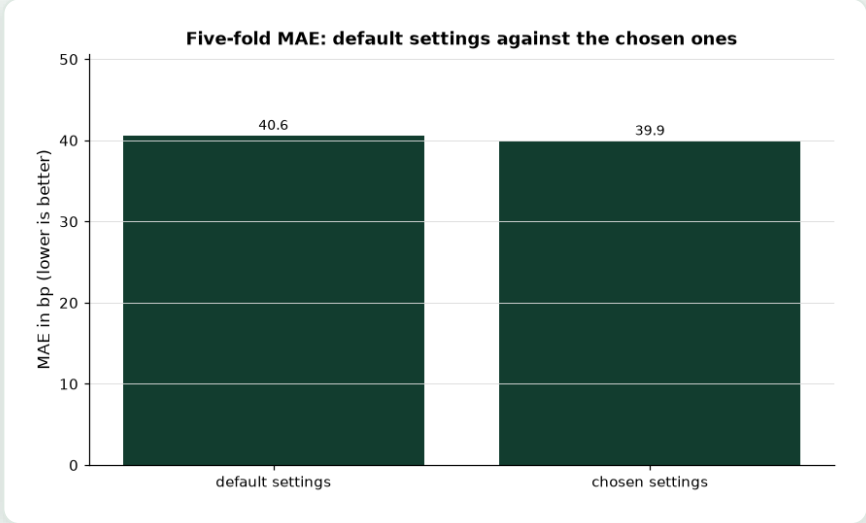
Setting	Chosen	OOB R2	MAE bp
n_estimators	500	0.8821	41.1
max_features	0.75	0.8851	40.7
min_samples_leaf	2	0.8833	41.3
max_depth	6	0.8863	40.4

The first panel is out-of-bag error against the number of trees: most of the gain is in the first hundred. Depth 2 is far too shallow; past a few bonds per leaf, bigger leaves lose.

Two settings together, and does tuning matter here



300 trees, five folds	R-sq.	MAE bp
default settings	0.8844	40.6
chosen settings	0.8872	39.9



The difference is smaller than the fold-to-fold spread of section 5.4. On this file, tuning the forest's settings moves the error by less than the noise.

Excel side by side

Excel side by side (tested in Excel)

`=EXP(C2)`	a fitted log spread in bp, filled down
`=AVERAGE(ABS(EXP(B2:B616)-EXP(C2:C616)))`	MAE 39.9947 bp
`=AVERAGE(F2:F6)`	mean of five folds 39.9947
`=STDEV.P(F2:F6)`	fold sd 11.6052, as printed
`=STDEV.S(F2:F6)`	12.9750: divides by 4, not printed
`=1-SUMXMY2(B2:B616,C2:C616)/DEVSQ(B2:B616)`	out-of-bag R-squared 0.8816

No regression tree, forest, or cross-validation in Excel. The scores are arithmetic, once the fitted values are pasted.

Save the week: git tag

1. Module

mlkit.py, ensemblekit.py and its tests

2. Random folds

results.csv: six rows, rows = cv

3. By issuer

six more rows, rows = cv_by_issuer

4. Tag

git tag week1

```
git diff week1~1 week1 -- results.csv
+ols,without coupon,cv_by_issuer,45.7623,0.8117
+tree_depth4,without coupon,cv_by_issuer,55.9545,0.7893
+forest_300,without coupon,cv_by_issuer,45.2195,0.8506
```

week1~1 is one commit before the tag: the diff shows the six rows by issuer. 21 tests pass at the end of today.

The AI check

The docstring is the prompt. The code that comes back runs, and its number looks fine.

1

Docstring

Which rows fit, which rows choose, which rows report: what you asked for.

→

2

Code back

Written for the course in the style of an AI answer. Read it.

→

3

Test first

Assert what the docstring promises. Run it: it fails.

→

4

Fix

Find the rule it broke, fix it, run the test again.

Today: the forest is cross-validated on all 821 bonds: Course 4's closed test bonds are scored, and the MAE (38.1 bp) looks fine.

Code that runs is not the same as code that is right.

What you can do now

- A regression tree's leaf is a mean; oob_score_ is R-squared
- Thresholds from out-of-fold probabilities on training rows
- Read the fold sd before the gap; compare fold by fold
- Give a forest its inputs in a form a tree can use
- Split by issuer for the question about new issuers

Week 2, Day 1 Boosting begins: AdaBoost